BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Object Type LHR.

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Note: Object Type LHR.

Note: (Group-i)

Q1. During nitration of benzene, the active nitrating agent is:

(A) NO3
(B) NO2
(C) NO2+
(D) HNO3

Q2. Chlorine heptaoxide reacts with water to form:

(A) HClO
(B) HClO4
(C) HClO3
(D) HClO2

Q3. Which of the following reagents will react with both aldehydes and ketones?

(A) Tollen's reagent
(B) Fehling's solution
(C) Grignard's reagent
(D) Benedict solution

Q4. Which ion will have maximum value of heat of hydration:

(A) Na+
(B) Cs+1
(C) Ba+2
(D) Mg+2

Q5. Which is not a calcarious material :

(A) Lime
(B) Clay
(C) Marble
(D) Marine shell

Q6. Select from the following which one is an alcohol:

(A) CH3-O-CH3
(B) CH3-CH2-OH
(C) CH3COOH
(D) CH3-CH2-Br

Q7. Laughing gas is chemically

(A) N2O4
(B) N2O2
(C) N2O
(D) NO

Q8. Mark the correct statement:

(A) All lanthanides are present in the same group
(B) All halogens are present in the same period
(C) All the alkali metals are present in the same group
(D) All the noble gases are present in the same period

Q9. Elimination bimolecular reactors [sic, print for 'reactions'] involve:

(A) First order kinetics
(B) Second order kinetics
(C) Third order kinetics
(D) Zero order kinetics

Q10. Which of the following is not a fatty acid:

(A) Propanoic acid
(B) Acetic acid
(C) Phthalic acid
(D) Butanoic acid

Q11. Formula of chloroform is:

(A) CH3Cl
(B) CCl4
(C) CH2Cl2
(D) CHCl3

Q12. Which halogen will react spontaneously with Au(s) produce Au3+ :

(A) Br2
(B) I2
(C) Cl2
(D) F2

Q13. Rectified spirit contains ethyl alcohol :

(A) 80%
(B) 85%
(C) 90%
(D) 95%

Q14. The strength of binding energy of transition elemer [sic, print for 'element'] depends upon:

(A) Number of electron pair
(B) Number of unpaired electrons
(C) Number of neutrons
(D) Number of protons

Q15. Which element forms an ion with charge +3:

(A) Be
(B) Al
(C) C
(D) Si

Q16. Ethanol can be converted into ethanoic acid by:

(A) Hydrogenation
(B) Oxidation
(C) Hydration
(D) Distillation

Q17. Ketones are prepared by the oxidation of:

(A) Primary alcohol
(B) Tertiary alcohol
(C) Secondary alcohol
(D) All of these

Essay Type

Time Allowed: 2.40 hours | **Maximum Marks:** 68

> Note: Essay Type

> Note: Group - I LHR.

Q2. (Marks 16) Write short answers to any EIGHT (8) questions: (8x2=16)

(i) Why the ionic radii of negative ions are larger than the size of their parent atoms?

(ii) Why the graphite is a good conductor?

(iii) Complete and balance the equations : (a) LiNO3 --Heat--> (b) NaNO3 --Heat-->

(iv) Why the aqueous solution of Na2CO3 is alkaline in nature?

(v) Give the chemical formula of : (a) Kaolin (b) Feldspar

(vi) Give the four difference of boron from other elements of group IIIA.

(vii) Give four uses of borax.

(viii) Give four similarities of sulphur and oxygen.

(ix) Give four uses of sulphuric acid.

(x) Mention any four qualities of a good fertilizer.

(xi) Give the composition of a good Portland cement.

(xii) Define the term 'setting of cement'. Also describe reactions taking place in first 24-hours.

Q3. (Marks 16) Write short answers to any EIGHT (8) questions: (8x2=16)

(i) Oxidizing power of halogen depends upon which factors.

(ii) What do you know about disproportionation reactions? Give example.

(iii) Give two uses of bleaching powder.

(iv) What are interstitial compounds?

(v) What is anode coating?

(vi) Give four examples of ortho-para directing groups.

(vii) Discuss catalytic oxidation of benzene.

(viii) What is iodoform test? Give its uses.

(ix) What do you know about silver mirror test?

(x) How would you prepare carboxylic acids from Grignard Reagents?

(xi) How would you prepare acid anhydride from acetic acid?

(xii) Why first four members of aliphatic acids are soluble in water?

Q4. (Marks 12) Write short answers to any SIX (6) questions: (6 x 2 = 12)

(i) What are homocylic [sic, print for 'homocyclic'] and heterocyclic compounds? Give one example of each.

(ii) Write the structural formulas of two possible isomers of C4H10.

(iii) How is methane converted to ethane?

(iv) Ozonolysis of alkene is used to locate the position of double bond, comment.

(v) Why is sigma bond inert?

(vi) Give four differences between S_N1 and S_N2 reactions.

(vii) How is tetraethyl lead prepared? Give its one application.

(viii) Ethanol reacts with conc. H2SO4 and gives different products at different temperatures, comment.

(ix) Write any two methods of preparation of phenol.

> Note: PART-I [sic, print for 'PART-II']

> Note: Note: Attempt any THREE questions.

Q5. (a) Discuss position of hydrogen with carbon family giving similarities and dissimilarities. (b) Write down any eight properties of lithium in which it behave differently from its group members.

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(b) Write down any eight properties of lithium in which it behave differently from its group members.

Q6. (a) How does HNO3(dil) react with: (i) Cu (ii) Zn (iii) Mg (iv) Sn (b) Describe following general characteristics of transition elements : (i) Binding energies (ii) Paramagnetism

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Q7. (a) Explain any four types of structural isomerism by giving one suitable example in each. (b) What is aldol condensation? Give its mechanism.

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Q8. (a) Explain acidic nature of alkynes in detail by giving two examples. (b) What are beta-elimination reactions? Differentiate between E1 and E2 reactions.

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(b) What is fermentation? How ethanol is prepared from starch?

GROUP – II

Objective Type LHR.

Time Allowed: 20 Min. | **Maximum Marks:** 17

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> Note: Objective Type LHR.

> Note: (Group-II; scan note box misprints 'Group-I')

Q1. Hydrogen bond is the strongest between the molecules of :

(A) HF
(B) HCl
(C) HBr
(D) HI

Q2. Formula of chloroform is :

(A) CH3Cl
(B) CCl4
(C) CH2Cl2
(D) CHCl3

Q3. Ketones are prepared by the oxidation of :

(A) Primary alcohol
(B) Secondary alcohol
(C) Tertiary alcohol
(D) All of these

Q4. Which is not a calcarious material:

(A) Lime
(B) Clay
(C) Marble
(D) Marine shell

Q5. Keeping in view the size of atoms, which order is the correct one:

(A) Mg > Sr
(B) Ba > Mg
(C) Lu > Ce
(D) Cl > I

Q6. Rectified spirit contains ethyl alcohol :

(A) 80%
(B) 85%
(C) 90%
(D) 95%

Q7. Tincal is a mineral of:

(A) Al
(B) B
(C) Si
(D) C

Q8. For which mechanism the first step involved is the same :

(A) E1 and E2
(B) E2 and S_N2
(C) S_N1 and S_N2
(D) E1 and S_N1

Q9. Which of the following is not a fatty acid?

(A) Propanoic acid
(B) Butyric acid
(C) Valeric acid
(D) Phthalic acid

Q10. Which one of the following is a typical transition metal:

(A) Sc
(B) Y
(C) Ra
(D) Co

Q11. Which of the following sulphates is not soluble in water:

(A) Sodium sulphate
(B) Potassium sulphate
(C) Zinc sulphate
(D) Barium sulphate

Q12. Linear shape is associated with which set of hybrid orbitals :

(A) sp
(B) sp2
(C) dsp
(D) dsp2

Q13. Which halogen occurs in a positive oxidation state :

(A) Fluorine
(B) Chlorine
(C) Bromine
(D) Iodine

Q14. The electrophile in aromatic sulphonation is :

(A) H2SO4
(B) HSO4
(C) SO3
(D) SO3+

Q15. Laughing gas is chemically :

(A) NO
(B) N2O
(C) NO2
(D) N2O5

Q16. The carbon atom of a carboxyl group is hybridized :

(A) sp
(B) sp2
(C) dsp
(D) dsp2

Q17. Which acid is used in the manufacture of synthetic fibre:

(A) Formic acid
(B) Oxalic acid
(C) Carbonic acid
(D) Acetic acid

Essay Type

Time Allowed: 2.40 hours | **Maximum Marks:** 68

> Note: Essay Type

> Note: Group II Lahore

Q2. (Marks 16) Write short answers to any EIGHT (8) questions: (8x2=16)

(i) What do you know about period 6 of the periodic table?

(ii) Why Na+ is smaller than Na atom?

(iii) What are alkali metals? Give name of alkali metals.

(iv) Give two difference between lithium and other alkali metals.

(v) Give chemical formulas of mica and bauxite.

(vi) How would you prepare borax from colemanite?

(vii) Give two uses of aluminium.

(viii) Draw structural formulas of dinitrogen pentoxide and dinitrogen oxide.

(ix) "Sulphuric acid is a dehydrating agent". Justify.

(x) Write different steps involved in the manufacturing of urea.

(xi) Why potasium [sic, print for 'potassium'] fertilizers are important for plants?

(xii) What reaction take place between 1 to 7 days during setting of cement?

Q3. (Marks 16) Write short answers to any EIGHT (8) questions: (8x2=16)

(i) Write any two uses of bleaching powder.

(ii) What is disproportionation reaction? Give one example.

(iii) Why HF is weaker acid than HCl ?

(iv) Why does damaged tin plated iron get rusted quickly?

(v) Give the prevention of metals from corrosion.

(vi) What are polycyclic aromatic hydrocarbons? Give examples.

(vii) What information do we get from X-ray study of benzene?

(viii) How does formaldehyde react with the following reagent : (a) HCN (b) NaHSO3

(ix) How will you distinguish between methanal and ethanal?

(x) Write any two reactions of carboxylic acids in which hydrogen atom of carboxylic group is involved.

(xi) What is meant by oxidative cleavage of alkenes? Give an example.

(xii) Write down the mechanism of ester formation.

Q4. (Marks 12) Write short answers to any SIX (6) questions: (6 x 2 = 12)

(i) Define catalytic cracking.

(ii) Define homocylic [sic, print for 'homocyclic'] and heterocyclic compounds.

(iii) Define hydrogenolysis. Give one example.

(iv) Why sigma bond is inert?

(v) How are cis and trans alkenes prepared from alkyne?

(vi) How are tetra ethyl lead and tetra methyl lead prepared?

(vii) Differentiate between electrophile and nucleophile.

(viii) What is denaturing of alcohol?

(ix) Write uses of ethanol.

> Note: PART-I [sic, print for 'PART-II']

> Note: Note: Attempt any THREE questions.

Q5. (a) Discuss the position of hydrogen in group I-A of the periodic table. (b) Describe with diagram the manufacture of sodium buy [sic, print for 'by'] Down's cell.

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Q6. (a) What is corrosion? How iron metal can be prevented from corrosion? (b) Give the reactions of HNO3(dil) with the following metals: (i) Hg (ii) Cu (iii) Sn (iv) Zn

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(b) Give the reactions of HNO3(dil) with the following metals: (i) Hg (ii) Cu (iii) Sn (iv) Zn

Q7. (a) Define organic compounds. Give any three features of organic compound. (b) Explain with mechanism the addition of sodium bisulphate to acetone.

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Q8. (a) Write note on polymerization of alkynes. (b) Give all methods for the preparation of alkyl halides.

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